Juggle Energy Minecraft Mod Idea

Overview

- A Minecraft mod focused on realistic energy generation, storage, and distribution.
- The mod introduces a power network that connects generators, batteries, control blocks, and powered devices.
- It is designed around the theme of industrial energy management, automation, and optimisation.
- The player can build systems that decide where power goes, when batteries charge, and which devices receive priority.

Core Concept

- The mod adds an energy system separate from vanilla redstone, but able to interact with it.
- Redstone devices can be powered through Juggle Energy if sufficient stored or generated energy is available.
- Each device requires a certain amount of power to activate or remain running.
- The system encourages players to manage supply, demand, storage, and efficiency.

Main Gameplay Loop

- Generate power using one or more sources.
- Store excess power in batteries.
- Distribute power to machines, redstone devices, and automation blocks.
- Use a control system to prioritise important loads and avoid power shortages.
- Expand the network with better generators, larger batteries, and smarter control logic.

Power Generators

Coal Generator

- Early-game and reliable power source.
- Burns coal or charcoal to produce energy.
- Produces steady output but requires fuel input.
- Suitable for basic setups and backup power.
- Could have upgrades for fuel efficiency or higher output.

Solar Power

- Generates power during daylight.
- No fuel cost, but dependent on weather and time of day.
- Best used with battery storage to smooth out daytime-only production.
- Could be upgraded for better output, faster charging, or reduced weather penalties.

Wind Turbine

- Generates power based on wind conditions and possibly height.
- Produces variable output depending on location and environment.

- Works well in open areas and high elevations.
- Good for renewable energy networks and passive generation.

Wave Power

- Generates power near water or ocean environments.
- Themed as a renewable coastal energy source.
- Could depend on biome, water type, or weather activity.
- Useful for base locations near oceans or large bodies of water.

Battery Storage

- Batteries store excess energy for later use.
- Different battery tiers provide increasing capacity and possibly better transfer rates.
- Batteries allow power networks to remain stable when generation drops.
- Stored power can be used at night, during storms, or when generators are offline.
- Higher-tier batteries could also be more efficient, faster to charge, or safer from overload.

Example Battery Tiers

- Basic Battery: small storage, early-game use.
- Advanced Battery: medium storage, better transfer rate.
- High-Capacity Battery: large storage for bigger networks.
- Industrial Battery: very high storage for late-game automation systems.

Control Block: Jbox

- The central control block of the system.
- Acts as the main interface for energy routing and automation rules.
- Lets players view generation, storage, demand, and active consumers.
- Can set priorities for devices and battery charging.
- Can enable or disable energy flow to specific systems.
- Can work as the main management block for an entire energy grid.

PLC Block Integration

- PLC blocks can connect to the Jbox for smarter control logic.
- PLCs can trigger events based on energy conditions.
- Example triggers:
- Turn on a generator when battery charge drops below a set level.
- Shut off non-essential machines during low-power conditions.
- Reserve power for critical systems.
- Switch between generators automatically.
- This adds an industrial automation feel and gives players more advanced control.

Redstone Device Interaction

- Redstone devices can be powered or activated through the energy system.
- Each device can have a defined energy cost.
- Devices only run when enough power is available.
- Devices may stop, slow down, or fail to activate if the network cannot meet demand.

- This creates a bridge between traditional Minecraft redstone and the new power system.

Dynamic Redstone Network Detection

- The system should automatically detect linked redstone circuits connected to the energy network.
- Power usage is calculated based on the total size and complexity of the redstone system.
- The more components in a circuit (repeaters, comparators, pistons, dispensers, etc.), the higher the energy requirement.
- Redstone wire length contributes to power cost, simulating transmission overhead.
- Large or complex automation builds require significantly more sustained power.
- Encourages efficient circuit design rather than excessive or redundant wiring.

Example Scaling Factors

- Short, simple circuits: low power consumption.
- Medium circuits with multiple components: moderate power consumption.
- Large automated systems with extensive wiring: high power consumption.
- Highly complex contraptions (e.g. farms, sorting systems, logic systems): very high power consumption.

Example Device Costs

- Doors, pistons, lamps, and simple mechanisms: low power cost.
- Dispensers, repeaters, rails, and moving systems: moderate power cost.
- Complex automated systems: high power cost.
- Large industrial machines: very high power cost.

Energy Management Features

- Priority system for deciding which devices stay online first.
- Load balancing across batteries and generators.
- Automatic switching between power sources.
- Warning states for low power or overloaded circuits.
- Optional manual override for player control.
- Energy dashboards showing production, consumption, and storage.

Progression and Balance

- Start with basic coal power and small batteries.
- Move into renewable sources and medium-tier storage.
- Unlock advanced control systems and high-capacity infrastructure.
- Late-game gameplay focuses on optimisation, automation, and efficient grid design.
- Balance should reward smart planning rather than simply placing more generators.

Possible Tiered Progression

- Early game: coal generator, basic battery, simple Jbox control.
- Mid game: solar and wind power, advanced batteries, basic PLC logic.
- Late game: wave power, industrial batteries, complex automation, full power routing control.

Visual and Thematic Direction

- Industrial, technical, and clean design language.
- Blocks should look functional and modular.
- Clear status indicators for power levels, charging, and active output.
- GUI should show simple but detailed energy data.
- Strong Juggle Energy branding across blocks and interfaces.

Player Benefits

- Adds a deeper automation system to Minecraft.
- Gives more purpose to power generation and storage.
- Lets players build realistic microgrid-style systems.
- Creates new technical challenges and engineering-style gameplay.
- Works well for players who enjoy automation, factory building, and optimisation.

Remote Connectivity System (4G & SIM Integration)

Overview

- Introduces remote control of energy systems via a simulated 4G network.
- Players can interact with their energy grid (via Jbox) from anywhere within network coverage.
- Adds a layer of infrastructure building focused on communication, not just power.

Core Components

SIM Card

- Crafted item required to enable network connectivity.
- Acts as an authentication and connection medium between player devices and Jbox systems.

Online Jbox

- Created by combining a standard Jbox with a SIM card.
- Gains remote connectivity capabilities.
- Can receive commands and send status data over the 4G network.

Player Device (Smart Controller)

- Handheld item used to interface with Online Jboxes.
- Allows players to:
- Trigger redstone events (e.g. pistons, switches).
- Enable or disable systems remotely.
- Monitor power usage, battery levels, and active devices.

4G Tower System

4G Tower (Base Station)

- Provides signal coverage in a defined radius.
- Both the player and the Online Jbox must be within signal coverage to communicate.
- Towers require a continuous power source to operate.

- Can be powered by batteries, solar panels, wind turbines, or other generators.
- If power is lost, the tower goes offline and network connectivity is interrupted.
- Encourages integration with Jbox systems to manage uptime and prioritise tower power.

Signal Coverage Mechanics

- Each tower has a circular coverage radius.
- Signal strength could degrade at the edge of the radius (optional feature).
- No connection is possible outside of coverage areas.

Network Expansion

- Multiple 4G towers can be placed to extend coverage.
- Towers can form chains or networks to cover large areas of the map.
- Encourages players to build infrastructure across their world.

Gameplay Mechanics

- Remote actions consume a small amount of energy from the network.
- Loss of power to towers disables communication.
- Encourages redundancy and backup systems for critical remote operations.
- Can integrate with PLC logic for advanced automation (e.g. remote overrides, alerts).

Example Use Cases

- Remotely activate a piston door from anywhere in range.
- Shut down non-essential systems during power shortages.
- Monitor base energy usage while exploring.
- Trigger automated farms or defenses remotely.

Expansion Ideas

- Power cables and transfer lines with loss over distance.
- Transformers for stepping power up or down.
- Fuses and breakers to protect circuits.
- Wireless monitoring or remote control through advanced blocks.
- Energy alarms and emergency shutoff systems.
- Integration with machines, farms, security systems, and transportation.

Summary

- Juggle Energy is a Minecraft mod built around energy generation, storage, and automated control.
- It adds realistic power management to redstone-based gameplay.
- The Jbox acts as the main control hub, with PLC blocks providing logic and automation.
- Multiple generator types and battery tiers create a progression system.
- The end result is a flexible industrial energy network that rewards planning and optimisation.